

File Settings Tools Debug Help

```
35 ?- house = house.  
true.  
36 ?- house = abc.  
false.  
37 ?- edge (a,b) = edge (C,b).  
ERROR: Syntax error: Operator expected  
ERROR: edge  
ERROR: ** here **  
ERROR: (a,b) = edge (C,b) .  
38 ?- edge(a,b) = edge(C,b).  
C = a.  
39 ?- edge(a,d) = edge(C,B).  
C = a,  
B = d.  
40 ?- edge(C,B) = edge(a,d).  
C = a,  
B = d.  
41 ?- edge(A,f(g,h)) = edge(a,B).  
A = a,  
B = f(g, h).  
42 ?- edge(A,A) = edge(b,c).  
false.  
43 ?- edge(A,A) = edge(b,b).  
A = b.  
44 ?- A = B, B = f.  
A = B, B = f.  
45 ?- A = B, B = f, A = e.  
false.  
46 ?-
```

Reason: Unification succeeds because both sides are identical atoms.

Reason: Unification fails because the two atoms are different constants.

Reason: Both sides are structures with the same functor (edge) and the same number of arguments (arity 2). Unification succeeds by instantiating the variable C to the atom a.

Reason: Unification succeeds by instantiating the variable C to a and the variable B to d.

Reason: Unification succeeds by instantiating the variable C to a and the variable B to d.

Reason: Unification succeeds. Variable A is instantiated to the atom a, and variable B is instantiated to the entire structure f(g,h).

Reason: Unification fails because of the shared variable rule. The variable A appears twice on the left, meaning both arguments must be identical. A cannot hold both b and c at the same time.

Reason: Unification succeeds. The shared variable A is instantiated to b, which correctly matches the right side where both arguments are b.

Reason: These are independent unifications separated by a comma (AND). Variable A is linked to variable B. and B is instantiated to f. Therefore. both A and B hold the value f.

Reason: Unification fails due to a logical contradiction. First, A and B are linked (A = B). Then B is instantiated to f, which forces A to also be f. The final goal attempts to unify A = e (which evaluates to f = e), and this fails because f and e are different constants.

Arwa Majed Alomar

9FY03

2220002864

Lab 4 – Task 1